

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

~~Quora~~

Chole $\rightleftharpoons$ Chole

MARKING SCHEME

JUNE 2016

CHEMISTRY 9189/5

1 Mark allocation for the table of weighing

- 1 mark for correct subtraction
- 1 mark for mass within the range 1.9 to 2.3 g excluding limits
- 1 mark for recording all values to the same decimal place

1.91 - 2.29 g

[Total: 3]

Mark allocation for the table of titration

- 1 mark for doing enough titrations *at least two with a diff of 0.20*
- 1 mark for all subtractions correct provided there are less than 2 zero starts.
- 1 mark for ticking precise titres, calculating average titre and completing the summary correctly *+ 2dp*

[Total: 3]

Mark allocation for accuracy

Calculate the difference as

$$\text{Diff} = \frac{1}{5} \left| \frac{\text{Titre}}{\text{mass (FA1) sup}} - \frac{\text{Titre}}{\text{mass (FA1) cand}} \right|$$

Then assign a mark using the accuracy and spread penalty table.

12

 total 18

accuracy marks spread penalty

diff	mark	diff	penalty
0.0050 0.010	12	0.20	0
0.0200	11	0.25	1
0.0350	10	0.30	2
0.0500	9	0.35	3
0.0750	8	0.40	4
0.1050	7	0.45	5
0.1150	6	0.50	6
0.1250	5	0.55	7
0.1350	4	0.60	8
0.1550	3	0.70	9
0.1850	2	0.80	10
0.2350	1	0.90	11
0.2351	0	1.00	12
		>1.00	12

(b) MnO_4^- (purple) is reduced to Mn^{2+} colourless;

[1]

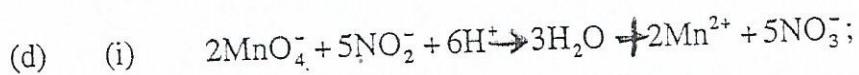
(c) (i) $C(\text{KMnO}_4) = \frac{1.5}{158} \text{ mol dm}^{-3}$

= $9.49 \times 10^{-3} \text{ mol dm}^{-3}$;

[1]

(ii) $n(\text{MnO}_4^-) = \frac{25 \times \text{ans to (c)(i)}}{1000}$

= $2.37 \times 10^{-4} \text{ mol}$; 0.000237 mol (3 s.fg) ecf. [1]



[1]

(ii) $n(\text{NO}_2^-) = \frac{5}{2} \text{ ans (c)(ii)}$; std form / (3 s.fg) ecf [1]

(iii) $C(\text{NO}_2^-) = \text{ans d(ii)} \times \frac{1000}{\text{titre}}$; std form / (3 s.fg) ecf [1]

6

$$(e) \quad n(\text{KNO}_2) = M_r(\text{KNO}_2) \times \text{ans d(i)} \times \frac{250}{1000} \quad [1]$$

$$\% \text{ of } \text{KNO}_2 \text{ in fertiliser} = \frac{n(\text{KNO}_2) \times 100\%}{m(\text{FA1})} \quad [1]$$

$$= \text{ans} \quad [1]$$

(f) NO_2^- is displaced by acid [1]

Qn 2 Mark allocation For the Design

Step I: use $m(\text{FA1}) \leq 0.4 \text{ g}$ / $\text{Vol}(\text{FA2}) \leq 50 \text{ cm}^3$ 1

Step II: displace NO_2^- by $50 \text{ cm}^3 \text{ HNO}_3$ which is XS 1

Step III: dilute to the mark of a $250 \text{ cm}^3 \text{ V}_f$ by $\text{dH}_2\text{O}(\text{FA3})$ 1+1

Step IV: Place it into a burette. 1

Step V: Remove any air bubbles 1

Step VI: Pipette 25 cm^3 of the 0.1 M NaOH(aq) into a Cf. 1

Step VII: Add 2 drops of Phenolphthalein 1

Step VIII: Titrate against FA3 to end point. 1

Step IX: Record the results in a suitable table. 1

Step X: Repeat 3 more times to reduce Errors. 1

Step XI: Calculate $n(\text{NaOH})$ in titrant as $\left(\frac{25 \times 0.1}{1000}\right)$ 1

Step XII: use the eqn: $\text{NO}_2^- + 2\text{H}^+ + \text{e}^- \rightarrow \text{NO} + \text{H}_2\text{O}$ 1

OR Rxn ratio 1:2 to deduce the $n(\text{NO}_2^-)$ that reacted with HNO_3 as:

$$n(\text{NO}_2^-) = \frac{1}{2} \left(\frac{\text{Vol}(\text{HNO}_3) \text{ used} \times 0.2}{1000} - \left(\frac{250}{\text{titre}} \right) n(\text{NaOH}) \right)$$

Step XIII: Calculate the % Composition as:

$$\frac{n(\text{NO}_2^-) \times M_r(\text{NO}_2^-) \times 100\%}{m(\text{FA1})}$$

[14 max 12]